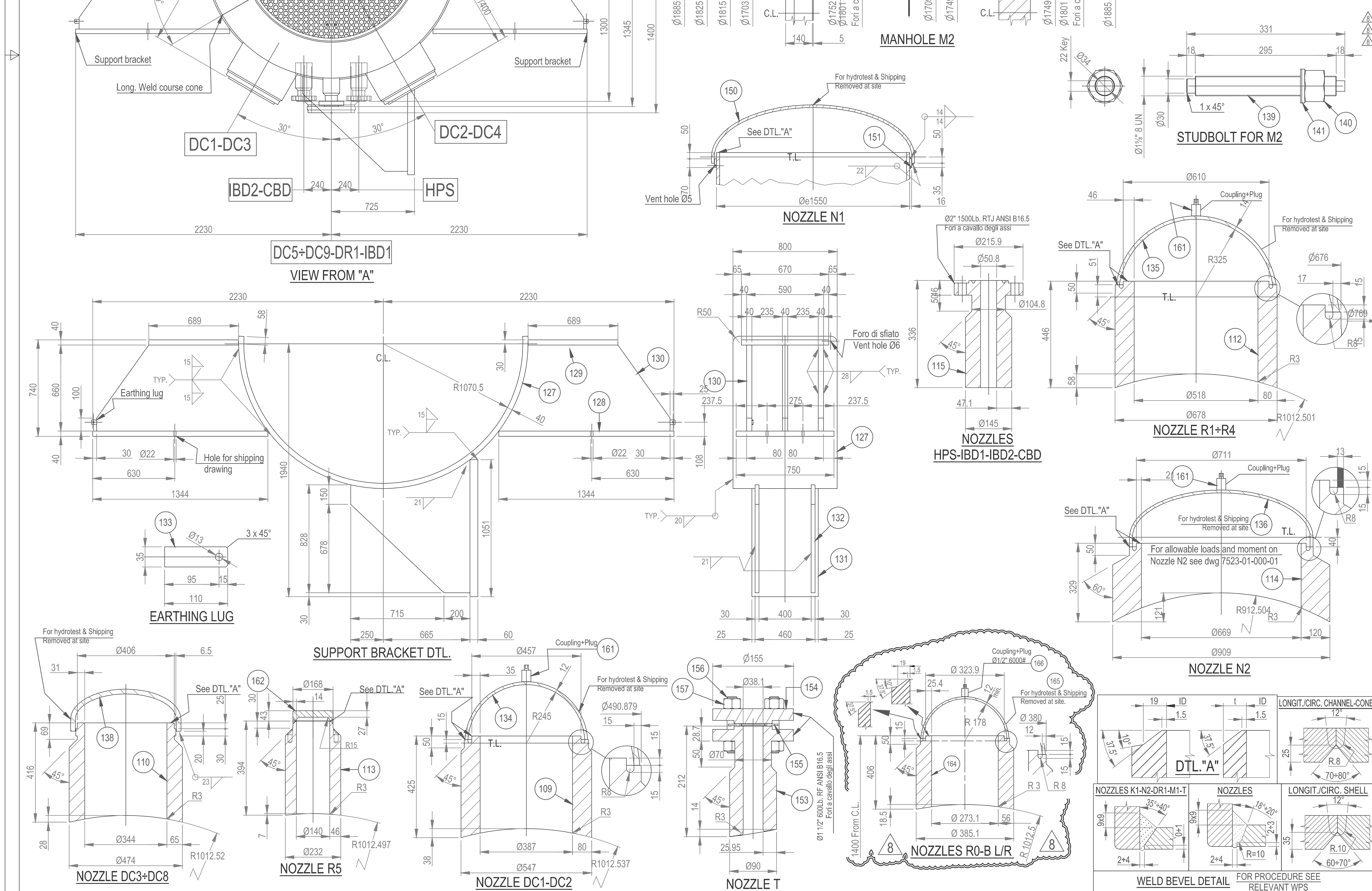

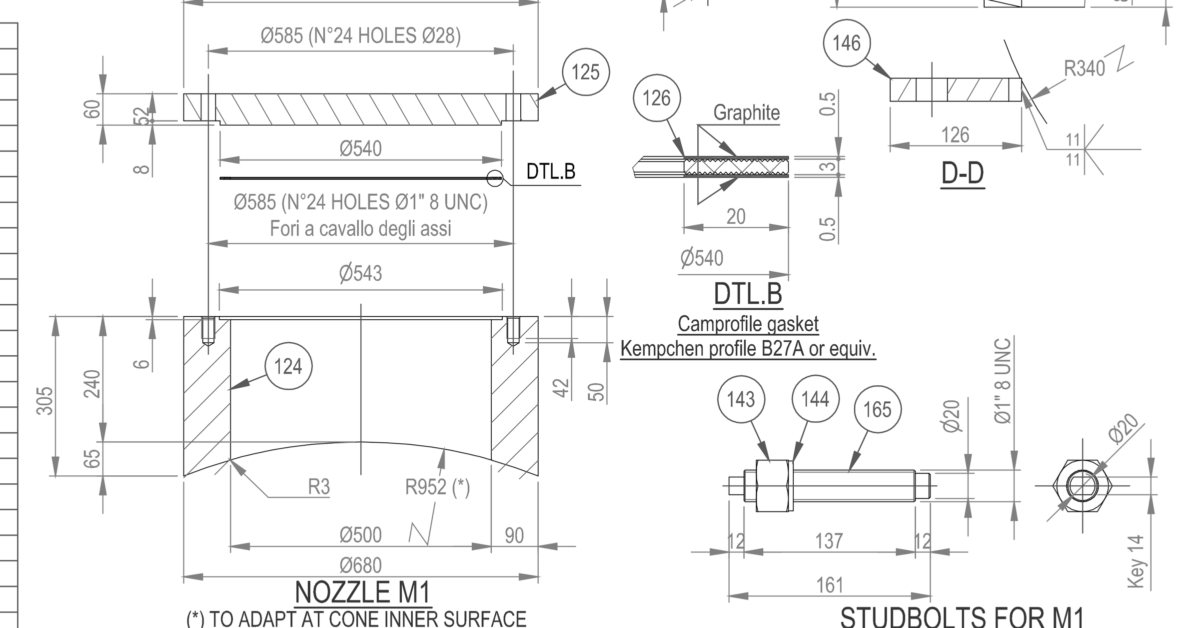

POS.	DESCRIZIONE/DESCRIPTION	DIMENSIONI/DIMENSIONS	MATERIALI/MATERIAL	Q.TY	CERTIFICATO CERTIFIED	SPARE PARTS	WEIGHT OF EACH PART kg.	NOTES
166 (△8)	Manicotto+Tappo/Coupling+Plug	Ø1/2" 6000# B1.20.1/B16.11	SA-105 N	2	2.2		1	
165 (△8)	Fondo emisferico/Hemisph. Head R0 B-L/R	Ri 178 x Thk.12 min.	SA-516 Gr.70N	2	3.1		66	
164 (△8)	Bocchello/Nozzle R0 B-L/R	Stub-End see dtl.	SA-508 Gr.3 Cl.2	2	3.1		364	
163	Fondo/Caps DC9	Øe168 Thk 43	SA-302 Gr B	1	3.1			
162	Fondo/Caps R5	Øe168 Thk 43	SA-302 Gr B	1	3.1			
161	Manicotto+Tappo/Coupling+Plug	Ø1/2" 6000# B1.20.1/B16.11	SA-105 N	7	2.2		0.35	
160	Golfare/Eyebolt	See dwg.	C.S.	1	2.2		1.84	
159	Supporto valvola sup./Valve support top	Øe54 x Lg.60	Alloy 602 CA	1	3.1		0.36	
158	Supporto valvola inf./Valve support bottom	Øe54 x Lg.155	Alloy 602 CA	1	3.1		2.22	
157	Dadi es./Hex. Nuts T	Ø3/4" 10 UNC-2B	SA-194 Gr.4	8	3.1	2	0.1	
156	Tiranti/Stud bolts T	Ø3/4" 10 UNC-2A x Lg.108	SA-193 Gr.B16	4	3.1	1	0.24	
155	Guarnizione/Gasket T	Ø1½" 600Lb. RF	S.W.316/Filled In/Out ring	1	2.2	2	0.16	
154	Flangia cieca/Blind flange T	Ø1½" 600Lb. RF	SA-182 F11 Cl.2	1	3.1		3.25	
153	Flangia/Flange T	Ø1½" 600Lb. RF see dtl.	SA-182 F11 Cl.2	1	3.1		9.16	
152	Flangia/Flange K1	See dtl.	SA-336 Gr.F11 Cl.3	1	3.1		43.34	
151	Piattina/Band N1	Øe1582 x Thk.16 Lg.70	SA-387 Gr.11 Cl.2	1	3.1		43.36	
150	Fondo Ell. 2:1/2:1 Elliptical Head N1	Øi1585 x Thk.22 min.	SA-516 Gr.70N	1	3.1		530.9	
149	Maniglia M1/Handle rod M1	Ø16 x Lg.270	SA 36	1	2.2		0.42	
148	Coppiglia/Cotter pin M1	Ø5 x Lg.60	C.S.	1	2.2		0.01	
147	Perno M1/Pivot M1	Ø26 x Lg.465 see dtl.	SA 36	1	2.2		1.9	
146	Cerniera Flg.Cieca M1/Hinge Blind flg. M1	126 x 100 x Thk.22	SA-387 Gr.11 Cl.2	2	3.1		1.35	
145	Cerniera M1/Hinge M1	139 x 115 x Thk.22	SA-387 Gr.11 Cl.2	2	3.1		1.54	
144	Rondella/Washer M1	Ø40/26 x Thk.4	HARDENED STEEL	24	2.2	3	0.02	
143	Dadi es./Hex. Nuts M1	Ø1" 8 UNC-2B	SA-194 Gr.4	26	3.1	3	0.22	
142	Tirante/Stud bolt M1	Ø1" 8 UNC-2A See dwg.	SA-193 Gr.B16	24	3.1	3	0.6	
141	Rondella cerca piano/Washer M2	Ø1⅝"	HARDENED STEEL	56	2.2	6	0.08	
140	Dadi es./Hex. Nuts M2	Ø1⅝" 8 UN-2B	SA-194 Gr.4	56	3.1	6	0.82	
139	Tiranti/Stud bolts M2	Ø1⅝" 8 UN-2A See dwg.	SA-193 Gr.B16	56	3.1	6	3.3	
138	Fondo Ell. 2:1/2:1 Elliptical Head DC3÷DC8	Øi422 x Thk.20 min.	SA-516 Gr.70N	6	3.1		43.91	
137	Fondello/Caps DC9	Ø6" x Sch.XS	SA-234 WPB	1	3.1		4.24	
136	Fondo Ell. 2:1/2:1 Elliptical Head N2	Øi744 x Thk.10 min.	SA-516 Gr.70N	1	3.1		67.84	
135	Fondo emisferico/Hemisph. Head R1÷R4	Øi650 x Thk.14 min.	SA-516 Gr.70N	4	3.1		64.55	
134	Fondo emisferico/Hemisph. Head DC1-DC2	Øi490 x Thk.12 min.	SA-516 Gr.70N	2	3.1		35.11	
133	Messa a terra/Earthing lug	110 x 35 x Thk.10	SA-516 Gr.70N	4	3.1		0.29	
132	Nervatura/Stiffening	1020 x 915 x Thk.30	SA-516 Gr.70N	4	3.1		126.95	
131	Lamiera centrale/Central plate	1051 x 510 x Thk.60	SA-387 Gr.11 Cl.2	2	3.1		253.01	
130	Nervatura/Stiffening	1314 x 630 x Thk.40	SA-516 Gr.70N	12	3.1		197.56	
129	Lamiera superiore/Top plate	670 x 690 x Thk.40	SA-516 Gr.70N	4	3.1		145.24	
128	Lamiera inferiore/Bottom plate	1345 x 750 x Thk.40	SA-516 Gr.70N	4	3.1		316.77	
127	Piastra di rinforzo/Reinf. plate	3500 x 800 x Thk.40	SA-533 Gr.B Cl.2	2	3.1		890.85	
126	Guarnizione/Gasket M1	Ø540/500 x Thk.4 m=2.25 y=20 MPa	Camprofile 316 S.S.+Graphit[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
125	Flangia cieca/Blind flange M1	Øe680 x Thk.60	SA-336 F11 Cl.3	[illegible]	[illegible]	[illegible]	[illegible]	
124	Bocchello/Nozzle M1	Ø680/500 x Lg.305 See dtl.	SA-336 F11 Cl.3	[illegible]	[illegible]	[illegible]	[illegible]	
123	Guarnizione distributore/Gasket channel M2	Ø1749/1709 x Thk.4 m=2.25 y=20 MPa	Camprofile 316 [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
122	Coperchio piano uscita/Flat cover outlet M2	Ø1885 x Thk.165	SA-336 F11 [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
121	Flangione uscita/Main flange outlet M2	Ø1885/1703 x Thk.145	SA-336 F11 [illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
120	Dadi es./Hex. Nuts DR1	Ø5/8" 11 UNC-2B	SA-194 Gr.4	[illegible]	[illegible]	[illegible]	[illegible]	
119	Tiranti/Stud bolts DR1	Ø5/8" 11 UNC-2A x Lg.108	SA-193 Gr.B[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
118	Guarnizione/Gasket DR1	Ø2" 600Lb. RF	S.W.316/Fille[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	
117	Flangia cieca/Blind flange DR1	Ø2" 600Lb. RF	SA-182 F11 Cl.2	[illegible]	[illegible]	[illegible]	[illegible]	
116	Flangia/Flange DR1	Ø2" 600Lb. RF see dtl.	SA-182 F11 Cl.2	[illegible]	[illegible]	[illegible]	[illegible]	
115	Flangia/Flange HPS-IBD1-IBD2-CBD	Ø2" 1500Lb. RTJ see dtl.	SA-508 Gr.3 Cl.2	4	3.1		[illegible]	
114	Bocchello/Nozzle N2	Stub-End see dtl.	SA-336 F11 Cl.3	1	3.1		477.44	
113	Bocchello/Nozzle R5	Stub-End see dtl.	SA-508 Gr.3 Cl.2	1	3.1		73.89	
112	Bocchello/Nozzle R1÷R4	Stub-End see dtl.	SA-508 Gr.3 Cl.2	4	3.1		473.38	
111	Bocchello/Nozzle DC9	Stub-End see dtl.	SA-508 Gr.3 Cl.2	1	3.1		39.41	
110	Bocchello/Nozzle DC3÷DC8	Stub-End see dtl.	SA-508 Gr.3 Cl.2	6	3.1		231.73	
109	Bocchello/Nozzle DC1-DC2	Stub-End see dtl.	SA-508 Gr.3 Cl.2	2	3.1		352.24	
108	Virola cono/Cone chamber	Øe2195/1550 x Thk.35 Lg.1973	SA-387 Gr.11 Cl.2	1	3.1		3090.88	
107	Virola distributore/Channel course VII	Øe1550 x Thk.25 Lg.800	SA-387 Gr.11 Cl.2	1	3.1		754.07	
106	Virola distributore/Channel course VI	Øe1885 x Thk.30 Lg.2000	SA-387 Gr.11 Cl.2	1	3.1		2588.32	
105	Virola mantello/Shell course V	Øe2141 x Thk.58 Lg.1258	SA-533 Gr.B Cl.2	1	3.1		3729.43	
104	Virola mantello/Shell course IV	Øe2141 x Thk.58 Lg.3100	SA-533 Gr.B Cl.2	1	3.1		9004.24	
103	Virola mantello/Shell course III	Øe2141 x Thk.58 Lg.3100	SA-533 Gr.B Cl.2	1	3.1		9004.24	
102	Virola mantello/Shell course II	Øe2141 x Thk.58 Lg.2700	SA-533 Gr.B Cl.2	1	3.1		7728.42	
101	Virola mantello/Shell course I	Øe2141 x Thk.58 Lg.2700	SA-533 Gr.B Cl.2	1	3.1		7512.66	

3.2	Materiale collaudato da ente ufficiale. Material tested by official authority.	MATERIALS TEST SYMBOL (Certification Acc. TO EN-10204) Simbolismo collaudo materiali
3.1	Materiale con certificato del produttore. Material with manufacturer certificate.	
2.2	Materiale senza certificato del produttore. Material without manufacturer certificate.	

- Il materiale elencato in distinta è da intendersi valido per N°1 apparecchio.
Quantity indicated on material list are for one shell.

LISTA MATERIALI / MATERIALS LIST

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
8	19/11/2014	Added Nozzles RO-B, Revised where marked & Reissued for Final Approval	Scotti	Picenni	Sarti
7	23/09/2014	Revised as indicated	Picenni	Sarti	Sarti
6	07/08/2014	Revised as indicated	Picenni	Sarti	Sarti
5	14/05/2014	Revised as indicated	Picenni	Sarti	Sarti
4	04/03/2014	Revised as indicated	Mazzoleni	Sarti	Sarti
3	25/10/2013	Revised as indicated	Mazzoleni	Sarti	Sarti
2	29/04/2013	General revised	Mazzoleni	Sarti	Sarti

ALFA LAVAL
SUISIO - BERGAMO - ITALY

CLIENTE / CUSTOMER } Fimco
ORDINE CLIENTE / PURCHASE ORDER } FZMC-13450-R1
IMPIANTO E LOC. / LOCATION & UNIT } Ammonia Urea Project
SERVIZIO / SERVICES } Sec. Reformer Effluent Waste Heat Boiler
TITOLO / TITLE } Construction drawing

SCALA/FORMATO NO / A1 — TIPO / TEMA / TYPE — NUMERO — SIGLA / ITEM No. — REV. 8

HEIDCO
By: seraj.mohammad
NO COMMENT
11:28 am, Jan 18, 2015

DISEGNO DEFINITIVO
Il cliente ha gia preso visione del disegno ed ha approvato le parti non evidenziate dall'ultima rev.
VALID FOR CONSTRUCTION
Drawing already approved by client except where indicated with last revision mark.
DATA: 19/11/2014 FIRMA: Sarti

REV.	ISSUE DATE	DESCRIPTION	DWN. BY	CHKD. BY	APPR. BY	APPR. BY (PROJ.)
7	[illegible]	[illegible]S INDICATED	SP	SS	SS	
6	07/08/14	REVISED AS INDICATED	SP	SS	SS	
5	14/05/14	REVISED AS INDICATED	SP	FG	SS	
4	05/03/14	REVISED AS INDICATED	PM	FG	SS	
3	25/10/13	REVISED AS INDICATED	PM	FG	SS	

OWNER: LORDEGAN PETROCHEMICAL COMPANY
Project: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR: HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER: HEIDCO Hampa Energy Engineering & Design Company
ALFA LAVAL SUISIO - BERGAMO - ITALY
TITLE: CONSTRUCTION DRAWING
Project No:2103 — HAMPA Doc. No. 7523-01-100-01 — Vendor Doc. No. 2103-A-VD-255-A1-MD-024 — REV. 8
Scale:
Sheet of

OWNER: LORDEGAN PETROCHEMICAL COMPANY
PROJECT TITLE: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR: HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER: HEIDCO Hampa Energy Engineering & Design Company
P.O. NO.: 2103-POR-A-HM-255 — ITEM NO.: 3-E-1401
REQ NO.: 2103-REQ-A-HM-255
DOC. NO. 2103-A-VD-255-A1-MD-024 — Rev. 8
NC – REVIEWED WITH NO COMMENT — DEPARTMENT:
C – DO NOT PROCEED WITH FABRICATION — DATE:
AC – APPROVED WITH MINOR COMMENTS — PERSON IN CHARGE:
RE – DOCUMENT WAS NOT ACCEPTED
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